Does outburst duration improve recurrence forecasts? A preregistered SS Cygni test and its measurement limits

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Abstract

Does the duration of a completed dwarf-nova outburst improve forecasts of the next onset beyond recent recurrence history? We tested this for SS Cygni with a century of archived visual observations, an outburst extractor and lognormal forecast clocks fitted on 1910–1959. After freezing the models and publishing the protocol, we opened 1960–2017. Adding duration raised the specified working interval score by 0.0165 nats per issuance (90% cluster interval 0.0049 to 0.0281; 441 issuances), meeting the predeclared criterion after a documented post-access parser repair. The scored intervals are fitted by the extractor, not known bounds on physical onset, so this is a working-score gain rather than an established likelihood for the physical event. The original later-period and long/short-class comparisons were inconclusive. Exploratory follow-ups retained a positive point-score sensitivity, but redesigned report-window and observed-clock comparisons were inconclusive, and rolling level adjustments reduced an analogous gain from 0.0184 to 0.0132 nats. To test generality, a prospectively frozen screen opened only the pre-1990 records of 10 catalogue-selected dwarf novae. None qualified for the planned multi-star test: under the same rules their 1980s records supported far fewer usable forecasts per confirmed outburst than SS Cyg's. No duration effect was fitted for those stars. The study establishes a reproducible working-score result for one star, extraction and baseline. It does not identify physical memory, and its generality remains untested.

1. Introduction

Dwarf-nova outbursts provide a long record of a repeating but irregular astrophysical process. A white dwarf accretes from a companion through a disc; thermal instability associated with hydrogen ionisation can switch the disc between cool and hot states (Lasota 2001; Hameury 2020). A completed outburst may therefore reveal something about the state from which the next cycle begins. The observational question is whether that information improves a forecast. Connecting an improvement to a particular physical state requires more than a correlation between measured durations and gaps.

SS Cygni has a long visual record in the American Association of Variable Star Observers (AAVSO) International Database. Bath and van Paradijs described a relation between outburst energy and the following waiting period (Bath and van Paradijs 1983). Cannizzo and Mattei (CM92) analysed 705 outbursts and showed why cycle definitions matter (Cannizzo and Mattei 1992). An onset-to-onset cycle already contains the duration of its first outburst. The cleaner duration–following-quiescence correlation was 0.223; duration also correlated with the preceding quiescence. Successive durations alternated, and a 12-day split separated long and short classes. CM92 interpreted successive quiet-interval correlations as possible secular variation. These associations, alternation and the class split are prior art, not discoveries of this study.

Hempelmann and Kurths (*Astronomy and Astrophysics*, 1990, 232, 356) examined dependence in symbolic outburst-width sequences. Searches for low-dimensional chaos found no attractor (Cannizzo and Goodings 1988). Cannizzo's disc calculations reproduced long/short alternation and slow changes in recurrence (Cannizzo 1993). Those results do not supply a held-out test of duration's contribution beyond a timing-history forecast. The closest forecasting precedent located in our bounded search is Grigorievskiy and colleagues' 2014 ITISE study, which tested point-onset forecasts on one V363 Lyr outburst. A separate search for short-term photometric precursors in SS Cyg found none (Price et al. 2007).

A forecast comparison must specify the information available at issue, the competing baseline and the observation being scored. Proper scoring rules evaluate whole predictive distributions (Gneiting and Raftery 2007), and temporal evaluation must preserve training order (Bürkner, Gabry, and Vehtari 2020). Applying those principles to visual outburst records is difficult: onset and end are inferred from noisy reports, observing can respond to the star, and gaps can hide events. The statistical interval returned by an extractor is not automatically an observed-data likelihood or a hard physical bound.

We report the original frozen comparison together with the research needed to establish what it measures. To our knowledge, it is the first preregistered, frozen-parameter probabilistic held-out evaluation of dwarf-nova recurrence forecasts, and the first held-out test of the completed outburst's duration beyond a specified timing-history baseline. That is an account of a bounded literature search, not proof of universal priority. The original calculation met its predeclared rule. Later review narrowed its interpretation to a working interval score, motivated different clocks and report windows, and led to a failed multi-star feasibility screen.

Keeping those results together avoids presenting measurement stability, a statistical decision and physical explanation as the same finding.

2. Evidence and design

2.1 Visual observations and extraction

The original analysis retained native Vis. observations with validation flag V. In the 1960–2017 evaluation interval, 249,356 of 500,479 deduplicated archive rows met that selector. Development used the earlier record; raw observations and observer identities are retained outside the public repository.

X4 extracts candidate outbursts at visual magnitude ten. It combines observer visits and fits local rising and fading crossings, allowing a shared level-error term. Confirmation requires bright evidence from distinct observers, or a qualifying single observer on separate nights. The crossing summaries are model-based intervals and points. Their construction, plug-in uncertainty and clipping do not establish a calibrated posterior or known containment of the latent crossing. A magnitude-ten crossing would itself remain distinct from the onset of a physical disc instability.

X4 passed a conditional synthetic recovery screen at this threshold and a seven-day coverage rule. A waveform family unseen during extractor development failed benchmark qualification before X4 recovery was scored; recovery on that family was not tested. Other threshold diagnostics failed. Qualification therefore did not establish complete real-event recovery or measurement validity across stars. The extraction and coverage choices define the comparison (Supplement S2.1).

2.2 Issuance, models and the working score

An observation-time end confirmation issues a forecast. Its predictors use only the preceding 400 days of reports: the completed outburst's fitted duration, the previous covered quiet gap and a recent mean of earlier covered log gaps. Future-row poisoning tests check this restriction. Historical upload and revision dates are unknown, so the analysis is an observation-time retrospective comparison, not an operational replay.

Each model is a lognormal quiet-gap clock conditioned on survival to the observed issue time. M0 is constant. M2 adds standardised log previous gap and recent gap level. M3 adds standardised log current duration with a nonnegative coefficient. M2 was fitted on 1910–1959; the duration offset was then fitted with M2 fixed and all parameters frozen. Its fitted coefficient was 0.0466 per development standard deviation. Routes without the required history use the same fallback in both arms and have exactly zero contrast.

The original working score assigns forecast probability to the next fitted X4 onset interval, or survival probability to a censoring cutoff. It is conditional on the clock's survival to issue. Coverage loss, supersession or record end can stop a forecast. These are calculations on extractor-defined evidence. Calling their contrast a proper score for physical onset would require a justified observation and selection model, which has not been supplied. Duration and the following quiet gap also share an estimated end; their errors need not be independent.

2.3 Frozen test and access record

The single confirmatory statistic was the mean M3−M2 working-score gain over all eligible evaluation issuances, including shared-fallback zeros. A small-sample-corrected cluster standard error used consecutive six-year blocks and Student's t. The predeclared criterion was a positive one-sided 95% lower bound. Era, type/within-type, calibration and robustness analyses were secondary and could not replace that test.

The protocol DOI, 10.5281/zenodo.24053338, predates the first evaluation request. That attempt failed before scoring because the archive returned literal None for missing uncertainty. The independently reviewed repair accepted that exact token without changing or dropping a row; no forecast uses that field. Rehearsal and invariance checks preceded continuation. The failed attempt remains preserved. This was a completed evaluation after a post-access format repair, not an untouched evaluation, and the DOI does not timestamp the repaired code.

2.4 Reassessment and transfer

After the result, an external review reopened the observation model. R1 audited censoring, rescored saved points and added endpoint errors to synthetic histories. R2b rebuilt outcomes from decisive-report windows and compared a fitted-end clock with an observed clock anchored to a decisive faint report. Its mark is the observed span of decisive brightness. This removes a literal interpolated end from that clock, but not shared reports, cadence-dependent anchor lag or selection. R2b also introduced windowed history baselines: B0 with recent-gap covariates, B1 with discounted history, and a preselected B2 state-space model. They are distinct from the original frozen M2.

All these SS Cyg analyses reuse opened data and are exploratory. Their specifications were fixed before their own fits, but they cannot become another confirmation. Report association and event completeness, physical containment or genuine censor survival, and a normalised coarsening-at-random observation kernel are separate premises. A report margin or timely visit does not prove them. Supplement S2.8 states the scoring consequences.

G3a was different: its rules were frozen before opening any of the candidate stars' records. It asked whether the pre-1990 visual data of the catalogue nominees and eclipse holds could support a later test. Qualification required supported timing-and-mark forecasts, eligible windows, stable nondegenerate unmarked B0 fits and no operational morphology flag; the holds also required an eclipse feasibility guard. It computed no duration–gap association or gain. No candidate test-period data were opened.

Table 1. What the evidence directly supports.

Input or comparison	Direct result	Further requirement for a stronger claim
Native visual reports and frozen X4	Extracted episodes, crossing summaries and coverage decisions	Validated event association and observation model for physical onset
Frozen M2/M3 and original access record	A held-out working-score contrast after the documented repair	Proper latent-onset scoring and adequate uncertainty assumptions
Exploratory R1/R2b	Sensitivity to outcomes, clocks, baselines and level adjustment	A new independent test of a fully specified observation/forecast rule
R2d and six-star simulations	Operating characteristics within declared synthetic processes	Transfer to real extraction, calendar yield and supported star dependence
Frozen G3a early-record screen	Qualification failure under the stated rules	No effect-size inference: no candidate duration effect was fitted

3. Results

3.1 Original working-score comparison

The predeclared criterion was met (Fig. 2): the mean working interval-score gain was 0.0165 nats over 441 issuances, with 90% cluster interval 0.0049 to 0.0281. This remains a result for the specified fitted intervals, extraction and measured-history baseline; a proper physical-onset likelihood is unestablished. Exponentiating gives a 1.7% increase in the geometric mean likelihood assigned to the scored evidence. It is neither an accuracy percentage nor an arithmetic probability gain. The original outcomes were 435 fitted onset intervals and six censored cases. Timing models applied on 436 issuances; five used the shared fallback. M2 itself gained 0.117 nats over M0.

The original qualifications remain beside this result. The 1993–2017 segment gave 0.0057 nats (−0.0107 to 0.0221); the predeclared class-plus-within-class rule gave 0.0025 (−0.0022 to 0.0072). Both were inconclusive. The earlier evaluation segment overlaps CM92's published association record, and no era-difference test was specified. Positive robustness variants do not supply independent confirmations.

Eight block means were positive, and leave-one-block-out means remained positive. Those checks do not validate the dependence model. The sample's predictive intervals overcover under the extraction-based coverage bounds; relative score gain does not establish calibrated physical-onset or operational forecasts (Supplement Fig. S1). Figure 1 illustrates one forecast selected by observation time, not score. Its model medians for remaining wait are 27.57 days for M2 and 26.77 for M3; it is an example, not an average scheduling benefit.

3.2 Measurement and clock sensitivities

The censoring audit found a coding rule requiring repair for future interval straddles, but the repair changed zero of the 441 original outcomes. R1's saved-point density sensitivity gained 0.0154 nats (0.0034 to 0.0274). These points remain uncertain extractions, not measured latent onset times. The proposed original report-interval sensitivity was largely unavailable: 434 associations lacked a finite lower clip. Its tiny supported subset could neither reproduce nor overturn the full result.

A separate corrected report-window construction yielded 818 available windows among 845 confirmed episodes, with 805 eligible and median width 1.15 days. Availability did not prove physical containment. R2b then scored the preselected state-space comparisons under those working-window rules (Fig. 3).

Exploratory contrast	Scored issuances	Mean gain, nats	Six-year 90% interval
H1: fitted-end clock, B2 duration	432	0.0075	−0.0008 to 0.0159
H2: observed clock, B2 bright span	431	0.0043	−0.0008 to 0.0094
H3: observed-clock gain minus matched fitted-end gain	431	−0.0031	−0.0078 to 0.0017

All intervals cross zero under six- and twelve-year blocking. There is no clear paired evidence that incremental gains differ between clocks. H3 is not an estimate of the causal contribution of end-time error. The delayed-clock population also differs from the original one: one coincident issue is represented once, and unsupported outcomes are excluded rather than assigned zero gain. The smaller estimates therefore cannot be read as a fraction of the original result surviving measurement correction.

In R1's separate endpoint experiment, added independent Gaussian end errors produced no positive mean gain across the tested null/history-dependent conditions. This limits that specified post-extraction error mechanism. Issuance and onset estimates were held fixed, large errors increased shared-fallback routing, and systematic or state-dependent observation errors were not tested. Neither this experiment nor the observed clock rules out a measurement explanation.

3.3 Baseline level and exploratory duration structure

The preselected B2 was not empirically the strongest comparator: its development likelihood was lower than B0 and B1, and its held-out base scores were descriptively lower. That motivates model review, not a unique diagnosis of the latent recurrence process. The measured gap level also shifted between eras. A frozen baseline can miscalibrate its level, and a predictor with a shifted mean can partially correct it.

Post-result diagnostics combined trailing mark centring with rolling location recalibration. The analogous point-B0 duration gain fell from 0.0184 to 0.0132 nats; the observing-rate gain fell from 0.0030 to 0.0014. Both the predictor and base changed, so these differences do not allocate causes or measure a component of C01. Guarded duration estimates stayed positive. Point-B0 retained positive lower limits; DF B0 and B2 intervals crossed zero. The observing-rate estimate also remained positive but inconclusive. Different significance labels do not establish a difference between predictors.

These historical diagnostics retained the original unguarded development coefficient and recalibrated from saved evaluation labels with proxy availability times. They were not an execution of the later proposed

rolling forecast algorithm. Trailing averages cannot remove arbitrary shared trends or states by construction. Observing-rate and duration contrasts therefore remain compatible with calibration and observation pathways.

The earlier post-result class fits also remain exploratory. A long/short indicator attained a larger in-sample likelihood than continuous log duration with historical nuisance parameters fixed and a free common level; adding within-class duration changed little. That supports considering a parsimonious class model, not exclusive attribution to class or an absence test. The frozen class rule's coding interacted with its fixed intercept; a shared level adjustment improved its descriptive score without rescuing the predeclared result. Full fits and alternation diagnostics are in Supplement S3.3–S3.5 and Fig. S2.

3.4 Precision planning and a failed transfer

The sequential synthetic planning study separated positive score moments from gain by the actual development-trained forecast. In a clean eight-star case with nominal individual latent-oracle gain 0.004 nats before measurement and learning, at 1,100 evaluation issuances, whole-star association-test rejection was 81.8%, but frozen-gain rejection was 36.65%. Dependence, missingness and delayed/noisy labels lowered performance; no robust sample-size guarantee followed. An oversized development coefficient could give a positive score moment and negative expected frozen gain. A six-star supplement likewise failed to establish a robust precision target. These are conditional point-Gaussian experiments, not window-score or archive-yield validation.

The frozen G3a screen opened only the pre-1990 records of 10 catalogue-selected stars. **None qualified** (Fig. 4). Under the frozen rules, only SS Aur exceeded the required 40 supported development issuances; it had 110 and an eligible-window fraction of 79.6%, but failed the operational long-bright-run screen. The other stars lacked sufficient supported issuances, sometimes also failing window or base-information gates. The eclipse holds used derived night streams; their synthetic no-eclipse baselines were inconclusive before the eclipse-profile tests were reached. Eclipses were not shown to cause those failures.

An independent reconstruction of SS Aur's 100.97-day flag found four groups of decisive-bright report nights, one represented by a single night. Among positive visit medians, 69.8% lay in the nondecisive threshold-bright band. These facts do not establish four physical outbursts, a plateau, a subtype or a photometric contamination mechanism.

The descriptive 1980s rate comparison is large: SS Cyg had 67 supported issuances and 73 confirmed episode peaks; the frozen candidate streams had 102 supported issuances and 485 peaks. These are issuance/peak rates, not paired conversion probabilities, and the candidates include derived eclipse-hold streams. In the separate ordinary raw-report diagnostic, attrition was largest at the requirement for a covered previous gap, including observing holes and ineligible onset windows. Assuming the frozen candidate rates continued, all ten stars ignoring qualification gates would project only 367.2 supported issuances over 36 years. This is a planning extrapolation, not an observed test count or a bound on other methods. We stop this paper's expansion here; no candidate duration effect was fitted and the proposed test period stays closed.

4. Discussion

4.1 A score result with an unresolved observation model

The original trained duration term improved the prespecified working interval score under its frozen extraction and history baseline. Independent numerical reconstruction and post-access repair audits make that computational result reviewable. They do not supply an observation model for the latent onset. The distinction changes the scientific claim: a probability assigned to a fitted interval can improve through properties of the reporting and extraction process as well as through better knowledge of the physical crossing.

R1 and R2b sharpen that limitation. Point rescoring is positive, but alternative report-window and observed-clock comparisons are inconclusive. Their different populations, clocks, marks and baselines prevent a clean measurement-error decomposition. Measured duration and the next gap share an end estimate; replacing that estimate with an observed report changes the coupling but retains cadence lag and selection. A mathematically valid score for coarsened observations would require the complete observation kernel and risk-set

premises, not merely narrow windows or decisive-looking reports.

Baseline calibration adds another route. The post-result guards reduce some gains, showing that recent-level information and mark centring matter. They do not establish how much of C01 is a level correction, and their own retrospective label-availability assumptions require qualification. A stronger state-space model cannot simply be assumed stronger: the particular B2 implementation fit development data less well. Those findings argue for reporting the model and observation contract with every forecast claim, rather than treating a duration coefficient as a property of the star alone.

Physical accounts remain possible. Cannizzo's calculations explain how unequal disc depletion can favour alternating outburst types; we found no reported prediction in that model of the following quiet-gap contrast after long versus short outbursts (Cannizzo 1993). Duration may also proxy a slowly changing disc or transfer state not captured by the fitted history. Alternation and a negative lagged-duration coefficient do not exclude a slower component. None of the present contrasts separates these mechanisms.

4.2 What the transfer failure means

G3a failed before a multi-star effect test existed. That is useful evidence about transport of this measurement and forecast contract. A dense-looking archive can contain few consecutive cycles with a covered previous gap, a decisive span and usable outcome. SS Cyg shows that the contract can be met: under the same chain its 1980s record supplied nearly one supported issuance per confirmed peak, against roughly one per five across the candidates. The single-star result rests on an unusually complete visual record, and a test of its generality needs records of comparable completeness. The prespecified thresholds, coverage rule, prefix history, morphology screen and eclipse reductions all helped define which opportunities counted. No observed effect in another star was rejected.

We retain the failure and stop expanding this paper rather than change rules until a cohort qualifies. This is a scope decision. It is not a theorem that a different measurement model, a later development split, later coverage or another archive could not support a multi-star analysis. Such work would need its own development and frozen evaluation; it could not turn this failed transfer into a pass.

Table 2. Assumptions and limits of the principal conclusions.

Calculated result	Required assumptions or design	Remaining limit
Original M3−M2 working-score gain	Frozen parameters, eligibility and cluster inference	Proper latent-onset likelihood and full measurement validity are unestablished
Positive saved-point sensitivity	Same fitted points and original associations	Points are uncertain extractions; no independent confirmation
Inconclusive R2b contrasts	Declared support, clocks and block summaries	Unsupported selection, containment/CAR and dependence remain assumptions
Smaller post-result guarded gains	Specified retrospective recalibration and centring	Neither a causal partition nor proof that all slow variation is removed
Synthetic association/gain performance	Declared Gaussian processes and star dependence	No universal effect, sample-size or real-window inference guarantee
No qualifying G3a cohort	Frozen early-record qualification and catalogue thresholds	A feasibility result for these rules; no candidate duration-effect inference

4.3 What would sharpen the answer

The most useful next evidence would independently connect report windows to the latent crossing and complete risk set, while making the information available at every historical update explicit. An independently qualified extraction would test whether the working-score result depends on the current measurements. A

new forecast comparison would then need a baseline whose adaptation, mark coding and nuisance estimation were all fixed together, with uncertainty checked under the admitted dependence and observation process.

The present study does not measure scheduling utility, operational calibration or a physical-memory parameter. SS Cyg observations from 2018 onward and the candidate stars' test records remain closed.

5. Conclusions

Adding completed-outburst duration met the frozen SS Cyg working-score criterion: 0.0165 nats per issuance, interval 0.0049 to 0.0281, across 441 issuances after the documented post-access format repair. The scored intervals have not been established as a proper observation likelihood for physical onset. The original later-period and class-rule secondaries and the redesigned headline comparisons remain inconclusive; level-adjustment diagnostics identify sensitivities without allocating causes. A prospectively frozen early-record transfer screen qualified no multi-star cohort. The resulting paper is a single-star working-score study and a documented measurement/transfer audit, not a demonstration of physical memory or general dwarf-nova predictability.

Methods in brief

Definitions and the original methods are in Supplement S1–S2; original and exploratory results, including report windows, precision studies and transfer attrition, are in S3. Review corrections are preserved in S4 and reproduction commands in S5. The claims ledger links each numerical statement to its evidence and limitations.

Acknowledgments

We acknowledge with thanks the variable star observations from the AAVSO International Database contributed by observers worldwide and used in this research.

[Placeholder: generative-AI declaration, to be written before submission. It will state the roles of GPT-6 Pro, Claude Opus 5.5 and GPT-6 Codex, and how their calculations, analyses and figures were tested and validated, as the journal requires.]

Data and code

The living repository https://github.com/BWBrook/dwarf-nova-memory contains protocols, source, frozen parameters, aggregate evidence, claims, figures and access receipts. Code is MIT licensed; text, figures and derived data are CC BY-SA 4.0. The original evaluation protocol is archived at https://doi.org/10.5281/zenodo.24053338.

AAVSO observations are available through the International Database. Raw observations, observer identities, event catalogues and individual forecast scores are not redistributed here. Reproduction uses the retained hashed inputs without another archive request (Supplement S5). Observer credit and co-authorship require review before release against the AAVSO guidelines.

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Fig. 1.

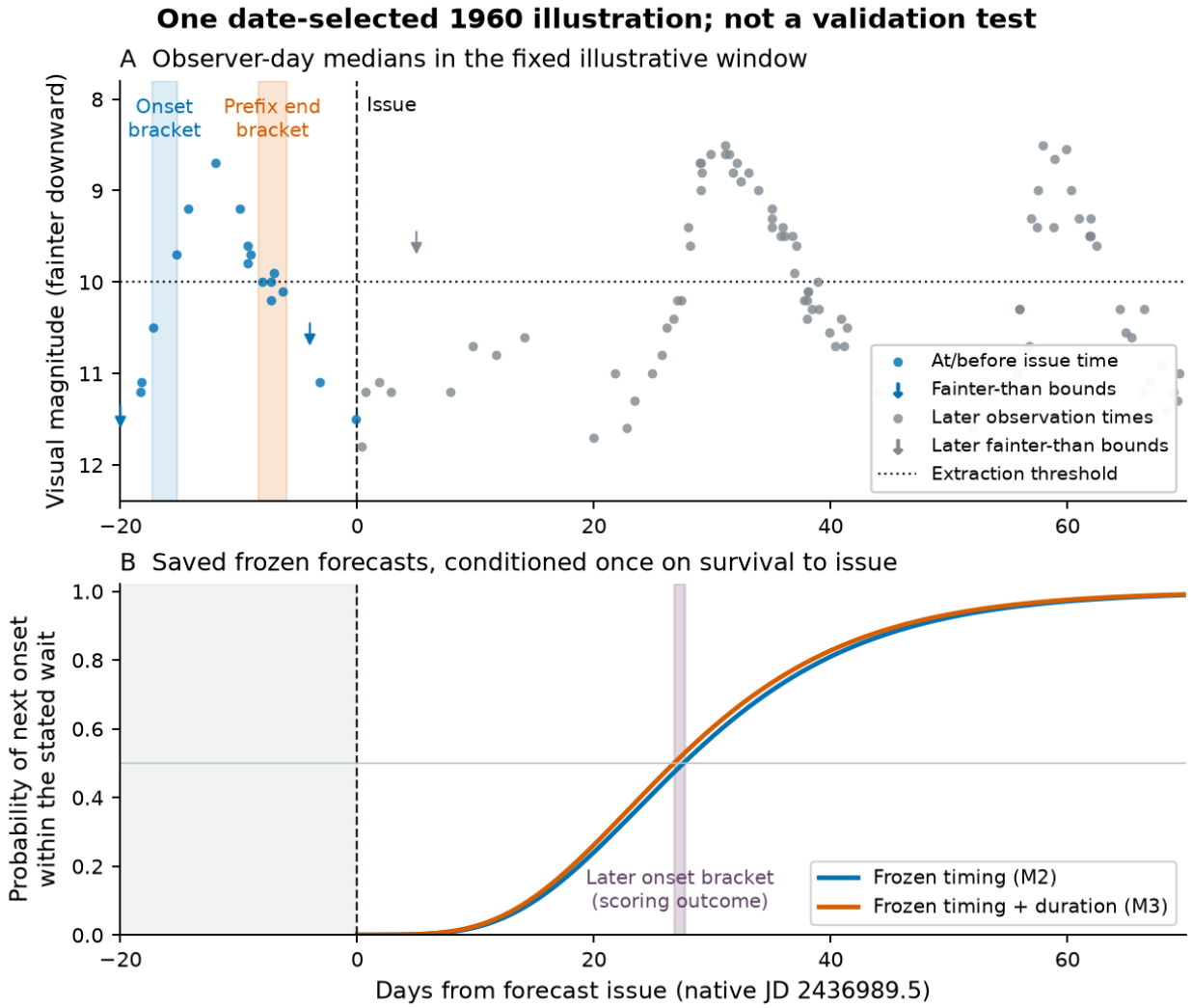


Figure 1: One forecast in the opened SS Cyg record, selected as the first eligible issuance by observation time rather than score. The upper panel shows observer-day display summaries, with detections separated from faint limits; arrows point toward fainter magnitudes. Prefix observations and the fitted crossing brackets define the completed-outburst duration available at issue. Display summaries are calculated separately on each side of issue. The lower panel shows the saved frozen timing and timing-plus-duration distributions, conditioned once on survival to issue. The later fitted onset interval supplies the working-score outcome; it is not a hard bound on physical onset. Reported observation times are known; historical upload and revision times are not. This example illustrates the retrospective information set, rather than validating the extraction or forecasting model. The completed comparison followed a post-access format repair.

306 **Fig. 2.**

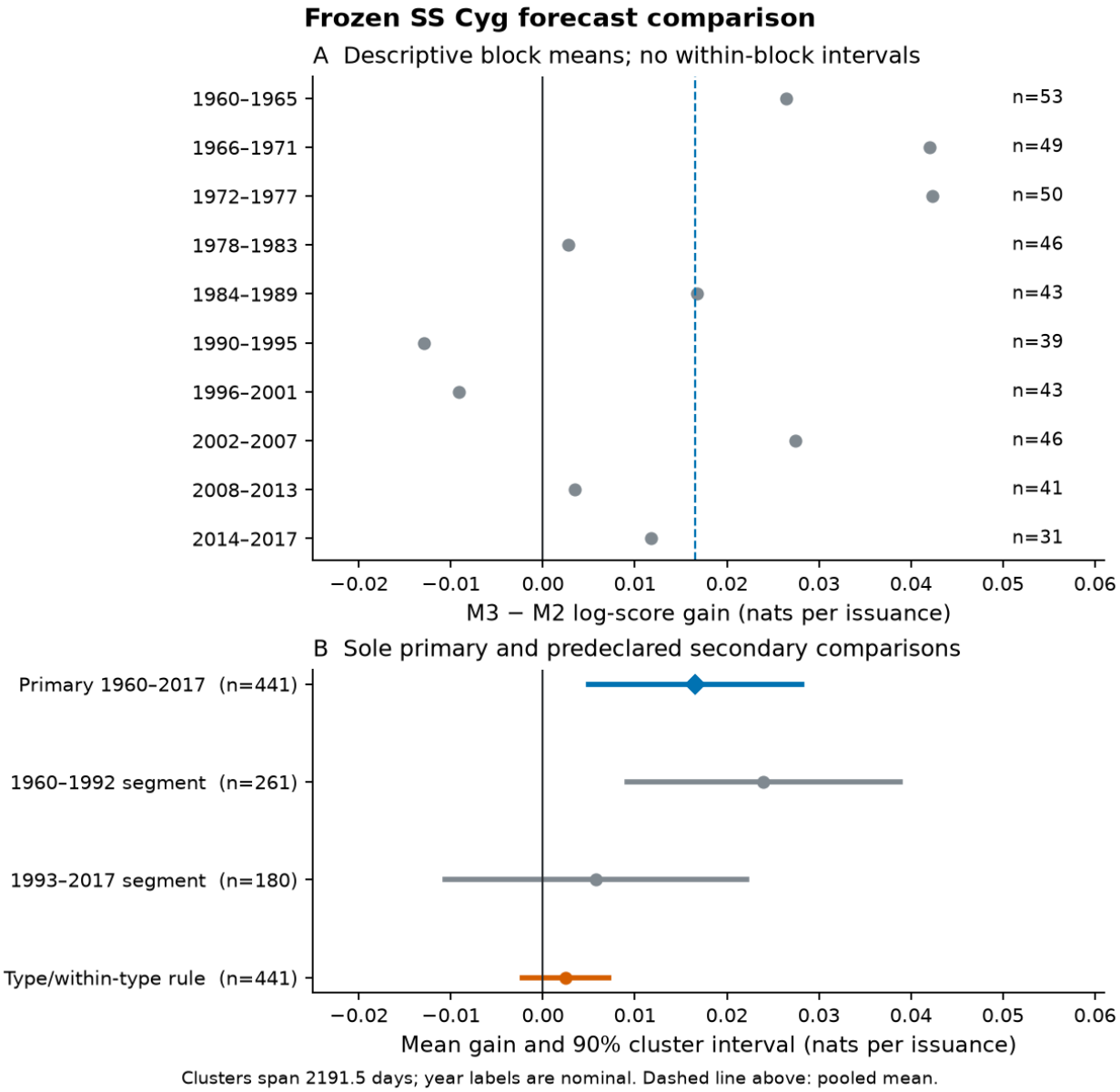


Figure 2: Frozen forecast gain and its predeclared qualifications. The primary paired working interval-score gain is 0.0165 nats per issuance, with interval [0.0049, 0.0281], over 441 forecasts. Horizontal intervals in the lower panel use the predeclared cluster procedure; only the primary comparison is confirmatory. The later segment and frozen type/within-type rule remain inconclusive. The upper panel shows descriptive means for the fixed blocks, without block-specific confidence intervals; calendar labels are approximate. The result concerns a gain in the stated working score beyond the specified timing baseline, following the recorded post-access format repair, and does not identify a physical mechanism.

307 **Fig. 3.**

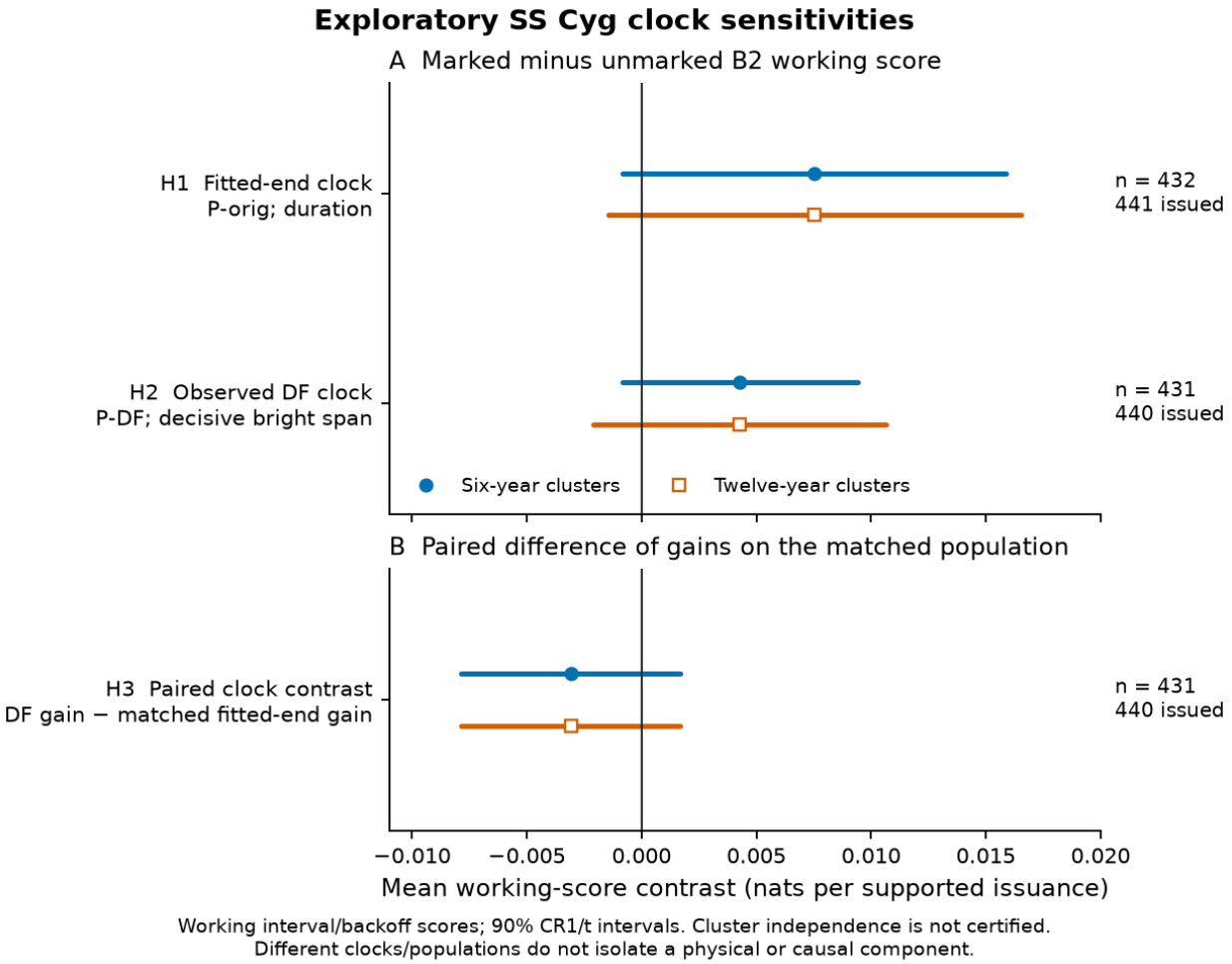


Figure 3: Exploratory SS Cyg comparisons after the external review. H1 compares marked and unmarked B2 models on the fitted-end clock; H2 uses the observed decisive-faint (DF) clock and a decisive-bright span. H3 is the paired difference of gains on the matched population, rather than a third duration effect. Circles and squares show the same estimates with six- and twelve-year CR1/t intervals. Every interval crosses zero. Counts distinguish scored from scheduled issuances. These working interval/backoff scores depend on the assumed observation windows; neither clock contrast nor retention identifies a measurement or physical cause. B2 was preselected, not empirically strongest. Generated by `manuscript/figures.py` from C55 aggregates.

Fig. 4.

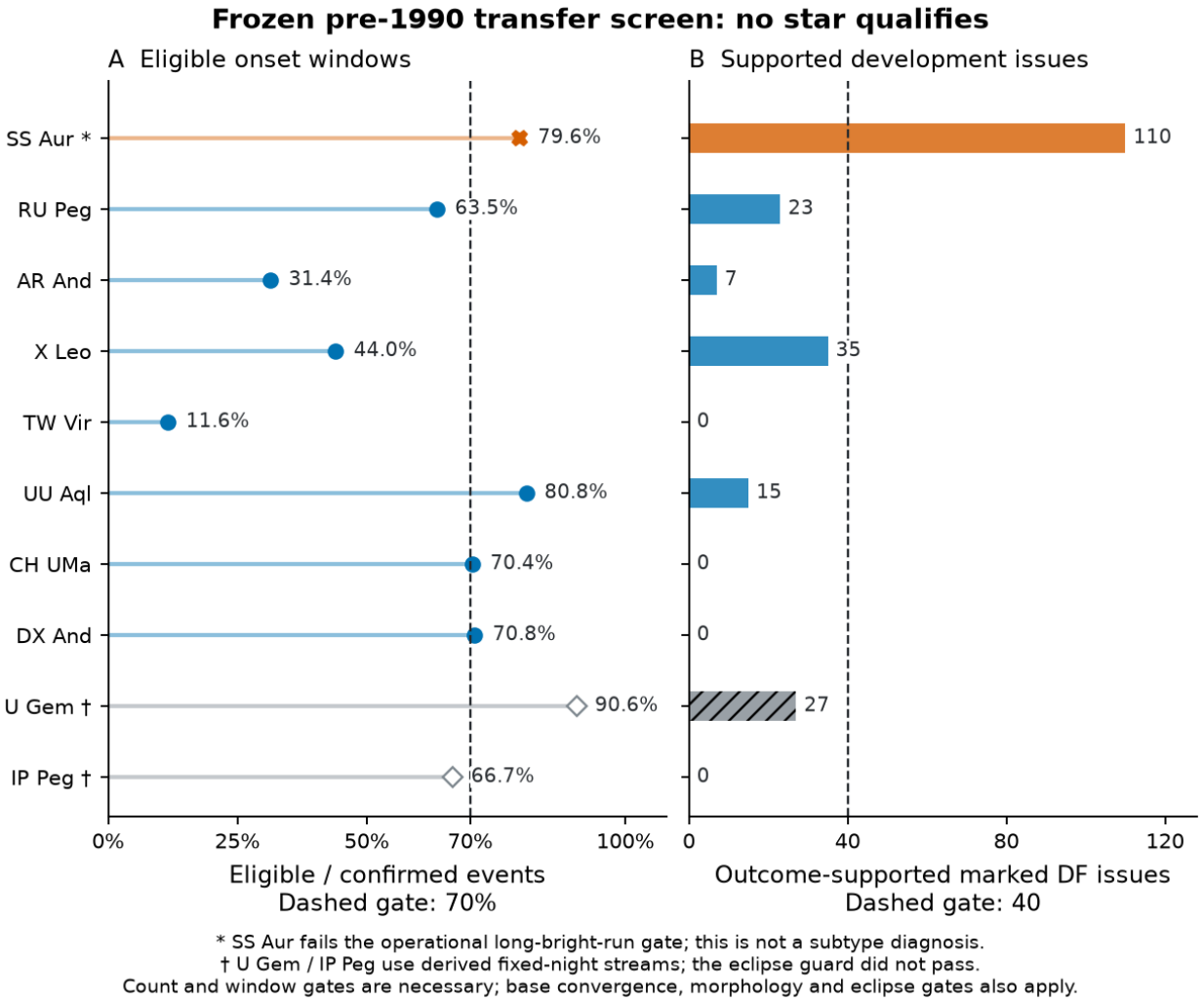


Figure 4: Prospectively frozen pre-1990 feasibility screen of ten catalogue-selected stars. The window fraction and supported-issuance count are necessary gates, shown against their frozen thresholds; baseline convergence, operational morphology and, for the two hold stars, eclipse handling also apply. No star passes every gate. SS Aur exceeds both displayed thresholds but fails the long-run flag; this does not diagnose its subtype or the cause of the report pattern. U Gem and IP Peg use the specified derived night streams, and their eclipse guard did not pass. No duration coefficient or forecast gain was fitted, and no later test records were opened. Generated by `manuscript/figures.py` from C68 aggregates.